# You Know More Than You Think

## Presentation Outline

### Audience

Mathematics majors who feel anxious about upper-division courses and students who are considering a master's degree or PhD in mathematics.

### Central message

Upper-division mathematics can feel daunting because the language, abstraction, and expectations change. That feeling does **not** mean that a mathematics major lacks mathematical ability or is starting from zero. Mathematics majors already have a substantial toolkit—algebra, logic, pattern recognition, functions, problem-solving, and persistence. Advanced courses ask students to connect, formalize, and extend that knowledge.

### Presentation goals

By the end of the presentation, students should be able to:

1. Recognize anxiety as a response to a challenging transition, not proof that they do not belong.
2. Identify the mathematical knowledge and habits they already bring to upper-division courses.
3. Use concrete strategies for learning definitions, proofs, abstraction, and notation.
4. Locate support through faculty, peers, tutoring, and study groups.
5. Understand master's and PhD study as possible pathways rather than tests of innate genius.
6. Replace self-blame with specific, evidence-informed actions.

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## Slide 1 — Mathematics Beyond the Comfort Zone

### Purpose

Establish the emotional focus of the presentation and reassure students immediately.

### Main points

- Upper-division mathematics may feel intimidating, especially when courses become proof-based and abstract.
- Students already know more mathematics than anxiety allows them to notice.
- The transition is not from “knowing no mathematics” to “knowing advanced mathematics.”
- It is a transition from using mathematical tools informally to making those tools explicit, connected, and rigorous.
- Graduate school is introduced as an option for students who are curious about continuing.

### Speaker emphasis

Open by naming the fear without reinforcing it: a difficult course can challenge a student's confidence while still being within that student's capacity to learn.

# Slide 2 — When the Rules of Math Change

## Purpose

Explain why upper-division mathematics feels different without describing students as deficient.

## Main points

- Upper-division courses ask students to explain *why* a statement is true, not only calculate an answer.
- Definitions become working tools.
- Examples are no longer the entire lesson; students must identify general structures.
- Proofs often take multiple attempts.
- Mathematical language and notation can make familiar reasoning feel unfamiliar.
- The concepts, algebra, logic, and problem-solving habits students already have remain active underneath the new language.

## Key reframing

| ANXIETY INTERPRETATION | MORE ACCURATE INTERPRETATION |
|---|---|
| “I do not know enough math.” | “I am learning a new way to use mathematics I already know.” |
| “Everyone else understands this immediately.” | “Everyone is learning how to make their reasoning more explicit.” |
| “I cannot start this proof.” | “I need to unpack the definitions and identify the first connection.” |

## Research connection

Moore (1994) identified understanding definitions, interpreting mathematical language, and knowing how to begin a proof as recurring difficulties in the transition to formal proof. These are teachable practices, not evidence that students have learned nothing before.

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# Slide 3 — The New Work of an Upper-Division Major

## Purpose

Make students' existing mathematical knowledge visible and show how upper-division courses extend it.

## Existing toolkit

- Algebraic manipulation
- Logical reasoning
- Pattern recognition
- Functions and relationships
- Spatial and quantitative reasoning
- Problem-solving and persistence
- Explaining an approach and checking whether it works

## New practices

1. **Definitions:** Rewrite a definition in your own words and create an example and a non-example.
2. **Proofs:** Identify the claim, evidence, and theorem or definition connecting each step.
3. **Abstraction:** Move from a concrete example to the structure that makes it general.
4. **Communication:** Use complete sentences to make mathematical reasoning visible.
5. **Revision:** Treat an incomplete proof as a draft rather than a verdict.
6. **Persistence:** Track better questions and clearer reasoning, not only immediate answers.

## Central reassurance

Students are not beginners at mathematics. They are becoming more deliberate about knowledge they have been building throughout the major.

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# Slide 4 — A Practical Study System

## Purpose

Give students an immediately usable routine for retrieving and extending the mathematics they already know.

## Before class

- Preview the definitions.
- Connect each new idea to a familiar example, theorem, algebraic move, or problem-solving habit.
- Mark unfamiliar notation instead of trying to memorize everything immediately.

## After class

- Reconstruct one example without looking.
- Explain one proof aloud or in writing.
- Name which earlier ideas make the new argument possible.

## When stuck

- Bring a specific attempt to office hours, tutoring, or a study group.
- State what you know and identify the new connection you cannot make yet.
- Treat a focused question as evidence of mathematical engagement, not failure.

## Key line

You do not need to feel confident before you begin. A routine can help you retrieve, connect, and extend the mathematics you already know.

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# Slide 5 — What the Research Supports

## Purpose

Provide research-backed evidence that anxiety, classroom structure, and learning habits affect mathematical performance.

## Evidence 1: Anxiety can consume cognitive resources

- Lyons and Beilock (2012) used fMRI research to study students with high mathematics anxiety.
- Greater activation in pain-related regions appeared while participants anticipated doing mathematics.
- The finding supports taking math anxiety seriously as a cognitive and emotional experience.
- It does not show that anxious students lack mathematical ability.

## Evidence 2: Active learning improves outcomes

- Freeman et al. (2014) synthesized 225 studies of undergraduate STEM courses.
- Active learning increased examination and concept-inventory performance by approximately 0.47 standard deviations compared with traditional lecture.
- Students in traditional lecture settings had higher odds of failing.
- The practical implication is to explain, retrieve, compare, discuss, and attempt mathematics—not only listen to it.

## Evidence 3: Mindset depends on context

- Yeager et al. (2019) conducted a large randomized study of a brief growth-mindset intervention.
- Effects were modest and strongest for lower-achieving students in schools where peers supported challenge-seeking.
- A positive message is not enough by itself; students also need a community and environment that make asking questions and attempting difficult work acceptable.

## Research caution

These findings describe patterns and supports. They do not predict any individual student's future, and they do not make graduate school the only successful path.

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# Slide 6 — Belonging Is Built Through Participation

## Purpose

Connect mathematical belonging to ordinary practices rather than confidence or performance perfection.

## Main points

- Mathematicians ask questions.
- They compare multiple approaches.
- They revise arguments.
- They get stuck and return to definitions.
- They learn from incomplete attempts.
- These are not signs that a student is behind; they are everyday practices of mathematics.

## Build a support network

**Faculty**

- Attend office hours before a crisis.
- Bring a specific attempt or question.
- Ask how to prepare for the next course or explore a field.

**Peers**

- Join or form a study group.
- Explain reasoning aloud.
- Compare definitions, examples, and proof strategies.
- Leave with individual understanding rather than copied answers.

**Self**

- Keep a proof or problem-solving journal.
- Record resolved questions and strategies that worked.
- Notice concepts that can now be explained more clearly.

## Key line

Students do not have to feel fearless to recognize how much they already know.

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# Slide 7 — What Graduate School Actually Asks

## Purpose

Make master's and PhD pathways more understandable and less mythologized.

## Master's degree

- Can provide deeper coursework and time to strengthen foundations.
- May help clarify interests in research, teaching, or applied mathematics.
- Can be one possible bridge between undergraduate study and later doctoral work.

## PhD

- Involves sustained training in producing new mathematics.
- Usually includes advanced coursework, qualifying milestones, seminars, advising, and research.
- Is an apprenticeship in a mathematical community, not a test of being a genius.
- Requires curiosity, preparation, persistence, communication, and mentorship.

## Preparation

- Analysis, algebra, topology, and advanced problem-solving courses can help.
- These courses build on mathematical habits students already have.
- No single course, exam, or transcript completely defines a student's potential.

## Mentorship

Students can ask faculty about:

- Fields of mathematics
- Reading courses
- Research experiences
- Summer opportunities
- Graduate program expectations
- What preparation would be most useful next

## Important distinction

Graduate school is an option, not an obligation. Mathematics majors can pursue teaching, statistics, computing, data, finance, public-sector work, and many other paths.

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# Slide 8 — A Plan for the Next Seven Days

## Purpose

Convert reassurance into a small, measurable action plan.

## One-week experiment

1. **Choose one definition.** Rewrite it in your own words and provide an example and a non-example.
2. **Attempt one proof.** Mark every definition, theorem, assumption, and connection you use.
3. **Ask one person.** Bring the attempt to office hours, tutoring, or a study group.
4. **Reflect briefly.** Record what changed in your understanding, not only whether the answer was correct.

## Key line

Confidence grows from repeated contact with hard ideas, supported by people who expect students to learn.

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# Slide 9 — Selected Sources

## Purpose

Show the evidence base and model responsible use of research.

## References


- Moore, R. C. (1994). “Making the transition to formal proof.” *Educational Studies in Mathematics, 27*(3), 249–266. https://doi.org/10.1007/BF01273731
- Lyons, I. M., & Beilock, S. L. (2012). “When Math Hurts: Math Anxiety Predicts Pain Network Activation in Anticipation of Doing Math.” *PLoS ONE, 7*(10), e48076. https://doi.org/10.1371/journal.pone.0048076
- Freeman, S., Eddy, S. L., McDonough, M., Smith, M. K., Okoroafor, N., Jordt, H., & Wenderoth, M. P. (2014). “Active learning increases student performance in science, engineering, and mathematics.” *Proceedings of the National Academy of Sciences, 111*(23), 8410–8415. https://doi.org/10.1073/pnas.1319030111

- Yeager, D. S., et al. (2019). “A national experiment reveals where a growth mindset improves achievement.” *Nature, 573*, 364–369. https://doi.org/10.1038/s41586-019-1466-y

## Research limitation

The studies support practical changes in how students interpret anxiety and learn mathematics. They do not determine whether any individual should pursue a master's degree or PhD.

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# Slide 10 — You Know More Than You Think

## Purpose

End with a direct, memorable reframing of mathematical identity.

## Closing message

- Upper-division mathematics can be daunting without being beyond a mathematics major.
- Anxiety can make existing knowledge difficult to retrieve or recognize.
- New notation does not erase familiar reasoning.
- Support, practice, and curiosity help students connect what they know to what comes next.

## Final contrast

**Replace:** “I do not know enough math.”

**Practice saying:** “I am connecting what I know to what comes next.”

## Final takeaway

Students belong in advanced mathematics not because every course feels easy, but because they already possess mathematical knowledge that can continue to grow.